

12 B Total energy E ,

$$E = \frac{1}{2}mv_0^2$$

$$= \frac{1}{2}m(\omega x_0)^2$$

$$= \frac{1}{2}m\left[\left(\frac{2\pi}{T}\right)x_0\right]^2$$

$$= \frac{2\pi^2mx_0^2}{T^2}$$

$$\frac{E'}{E} = \left(\frac{x_0'}{x_0}\right)^2 \left(\frac{T}{T'}\right)^2$$

$$E' = \left(\frac{x_0'}{x_0}\right)^2 \left(\frac{T}{T'}\right)^2 E$$

$$= (2)^2 \left(\frac{1}{3}\right)^2 (18) \quad (\because x_0' = 2x_0 \text{ and } T' = 3T)$$

$$= 8.0 \text{ mJ}$$

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